Energy sector:

The mitigation plan envisages the implementation of a comprehensive programme comprising 36 measures. These will be structured around six main components:

1. The first component will target the oil and gas sector through the National Oil Corporation (NOC) Mubādara 2030 initiative, which is part of the Global Methane Pledge. and will comprise 22 measures focusing mainly on reducing GHG emissions from flaring at oil and gas sites, with an additional 23th and final measure in the form of the implementation of an MRV system to monitor the programme's progress.

   The NOC and its subsidiaries companies acknowledge the value of advanced monitoring and alert systems in strengthening efforts to manage methane emissions within the oil and gas sector. In this regard, the Corporation has benefited from the reports and alerts generated by the Methane Alert and Response System (MARS), developed under the United Nations Environment Programme and the International Methane Emissions Observatory. These alerts provide timely and science-based insights that allow NOC and its subsidiaries to identify potential sources of emissions and prioritize areas for further assessment.

   Building on this information, the NOC is committed to following up on the alerts received, verifying their underlying causes, and working within available capacities to implement appropriate measures aimed at reducing or preventing methane emissions. This process of integrating the global monitoring tools into national practice, enhances both institutional readiness and sector-wide responsiveness.

   The NOC and its subsidiaries attach great importance to reducing methane emissions as a key area of interest in supporting national climate action. Recognizing the critical role of methane mitigation in achieving near-term climate benefits, the Corporation is closely following the work of the Oil and Gas Methane Partnership (OGMP), an international initiative dedicated to advancing methane emission measurement, reporting, and reduction in the oil and gas sector.

   By drawing on the principles and approaches promoted by the OGMP, the NOC seeks to explore opportunities to enhance transparency, learn from global best practices, and improve methodologies for monitoring, reporting, and managing methane emissions, in line with available resources and capacities. This interest reflects the NOC's broader effort to strengthen institutional capabilities and to foster a culture of accountability and continuous improvement across the sector.

The NOC's attention to such international initiatives is consistent with the objectives of the country's NDC, particularly in the oil and gas sector, where methane abatement represents one of the most cost-effective and impactful pathways for reducing greenhouse gas emissions.

By engaging in international initiatives such as MARS, and showing interest in international partnerships, the NOC aligns with the best global practices while tailoring its efforts to national circumstances and operational realities. Such engagement will be pursued, allowing to directly support the objectives of the NDC by contributing to improved data quality, enhanced transparency, and strengthened mitigation action in the oil and gas sector. In this way, NOC underscores its determination and commitment to explore every feasible opportunity to limit methane emissions and to progress towards a sustainable and low-emission energy future.

2. The second component will cover the electricity sector through a comprehensive programme comprising a package of energy efficiency measures targeting electricity consumption in each sector, with an overall objective of improving energy performance by 40% in street lighting and by 15% in other sectors (households, public services, commerce, agriculture and industry). These objectives would be achievable through "technical" measures such as the widespread use of smart metering systems, the rehabilitation and retrofitting of street lighting networks and full retrofitting to LED, the promotion of LED lighting in all sectors (residential, services, industry, etc.), the implementation of energy labelling and MEPS for electrical appliances in all sectors (air conditioning, refrigeration, electrical water heaters, washing machines, motors, pumps, generalization of ISO50001 where relevant, etc.), or smart mechanisms such as transferring part of energy subsidies to support energy efficiency and renewable energy in all sectors, or credit systems via energy bills to implement technical measures. All of this will be implemented with a win-win approach for consumers, energy producers and the Libyan state.

3. The third component will also target the electricity sector, from the perspective of electricity supply, and will mainly include three types of actions: (i) Grid modernization, including upgrades to transmission infrastructure and distribution networks, to reduce technical losses, (ii) Switch to gas (natural gas currently 67%,

will reach 85% in 2035) and, (iii) Promote energy efficiency in electricity production through the use of combined cycle power plants (currently 45%, will reach 90% in 2035).

4. The fourth component will also target the electricity sector, through the launch of a major programme to develop electricity production from renewable energies, with the aim of achieving a 22% share of the electricity mix through the main renewable technologies by 2035. This component will also include the launch of a strategic study and the commencement of the operational phase of the Green Hydrogen Technology Development programme.

5. The fifth component will cover Energy Efficiency, mainly the thermal energy uses by the economic sectors. This includes: (i) Introducing compulsory Energy Auditing programmes in the major consumer sectors and entities, (ii) Implementing an ambitious program to developing Cogeneration and Trigeneration where relevant, (iii) Promoting the optimization of process-heat consumption in the industries, and (iv) Promoting the optimization of Insulation of heating and cooling circuits

6. The sixth component under the energy umbrella will target the transport sector through the launch of a major programme focusing on four areas: (i) Enhance the National Mobility System by building capacity for public/mass transport, including improvements to the bus system (introduction of dedicated BRT lanes, modern high-capacity buses, real-time ITS signals, discounted tickets for students and those with limited incomes, and electronic ticketing in Tripoli), etc. (ii) Launch a national campaign to encourage mass transport involving the private sector, (iii) Promote Hybrid and Electrical vehicles (target : 100.000 vehicles in use in 2035), both covering both private and public sectors, in particularly for the latter through a "Green cars in government institutions" achieving a shift towards low-emission transport by gradually replacing the government vehicle fleet with electric or hybrid vehicles, (iv) Promote EE + "MEPS" + Maximum age of imported vehicles+ promoting driving practices + maintenance practices, etc.

Industrial processes :

The mitigation plan for the Industrial processes will target the three main sources of emissions, comprising three measures: (i) Ratify the Kigali Amendment to reduce HFC emissions, (ii) Reduce emissions from reheating furnaces (calcination, billet rolling, profile rolling and hot rolling) in the Iron&steel industry, (iii) Reduce the clinker/cement ratio from 0.93 to 0.90% by 2035, in particular by replacing clinker with blast furnace slag and fly ash from steelworks.

Agriculture, Forestry and Land Use Change:

The mitigation plan primarily envisages two actions aimed at strengthening the carbon absorption capacities of ecosystems in Libya, through two main programmes**:**

- Prime minister initiative: Planting 100 million trees, covering 150,000 hectares in various relevant regions and areas in Libya, to enhance carbon sequestration. This will include drought-resistant tree species around cities to act as windbreaks and microclimate stabilizers
- Ministry of agriculture initiative: Planting 10 million trees, targeting drought-resistant and easy adaptative tree species under Libya ecological context, covering 33,000 hectares in various relevant regions and areas in Libya, to enhance carbon sequestration while contributing to ecosystem preservation.

Solid Waste and Wastewater:

- The mitigation plan will cover both solid waste and wastewater treatment. It will include six main measures:
- Rehabilitation and retrofit of 4 among the largest uncontrolled landfills (Tripoli, Benghazi, Misrata and Sebha) and installation of biogas extraction networks and Flaring utilities
- Implementing 4 new sanitary controlled landfills (Tripoli, Benghazi, Misrata and Sebha) with CH_4 capture network and power generation utilities
- Implementing 4 large-composting facilities
- Implementing 4 Mecano-Biologic Treatment (MBT) facilities
- Rehabilitation and retrofitting of all Wastewater facilities
- Generalization of aerobic treatment processes in the Wastewater facilities

Carbon Intensity and NDC Targets 2035:

Carbon intensity allows a country to show that it is taking steps to decarbonise its modes of production even though its emissions may still be increasing due to economic growth. For instance, if the same mitigation measures were calculated using absolute emissions then Libya's unconditionally commitment would be to ***increase*** emissions by 20% relatively to 2023 levels by 2035. Emissions would decrease by 15% under conditional scenarios. Similarly, if the energy sector alone is examined, conditional mitigation measures would decrease emissions by only 13% under absolute emissions. If these same measures are recalculated using carbon intensity the reduction would be 47%.

Annex 3: GHG inventory of Libya between 2017-2023

The following table provides the GHG inventory for the period 2017-2023.

Categories	2017	2018	2019	2020	2021	2022	2023
Total National Emissions and Removals	77.109	79.915	84.314	78.530	88.892	91.427	97.324
1 - Energy	67.534	70.317	74.494	69.087	78.504	80.656	85.978
1.A - Fuel Combustion Activities	45.431	45.676	47.548	45.126	51.749	55.988	58.950
1.A.1 - Energy Industries	23.296	24.048	24.825	23.427	26.089	27.760	32.539
1.A.2 - Manufacturing Industries and Construction	1.509	1.640	1.794	1.570	2.273	2.070	1.980
1.A.3 - Transport	19.795	19.149	20.072	19.293	22.453	25.193	23.403
1.A.4 - Other Sectors	831	840	856	836	933	965	1.029
1.B - Fugitive emissions from fuels	22.103	24.640	26.946	23.961	26.755	24.668	27.028
1.C - Carbon dioxide Transport and Storage	NO	NO	NO	NO	NO	NO	NO
2 - Industrial Processes and Product Use	5.203	5.100	5.307	4.918	5.845	6.144	6.658
2.A - Mineral Industry	1.244	1.367	1.281	1.331	1.524	1.698	1.510
2.A.1 - Cement production	1.064	1.202	1.034	1.106	1.277	1.442	1.246
2.A.2 - Lime production	180	165	248	225	248	256	264
2.B - Chemical Industry	683	425	391	93	679	612	711
2.B.1 - Ammonia Production	590	417	274	72	604	612	678
2.B.8 - Petrochemical and Carbon Black Production	94	8	117	21	76	0	33
2.C - Metal Industry	806	811	1.135	1.012	1.169	1.326	2.063
2.C.1 - Iron and Steel Production	806	811	1.135	1.012	1.169	1.326	2.063
2.D - Non-Energy Products from Fuels and Solvent Use	35	39	32	22	29	27	27
2.F - Product Uses as Substitutes for Ozone Depleting Substances	2.334	2.334	2.334	2.334	2.334	2.334	2.334
2.F.1 - Refrigeration and Air Conditioning	2.334	2.334	2.334	2.334	2.334	2.334	2.334

3 - Agriculture, Forestry, and Other Land Use	3.326	3.408	3.411	3.382	3.361	3.406	3.430
3.A - Livestock	2.281	2.366	2.426	2.354	2.348	2.373	2.409
3.A.1 - Enteric Fermentation	1.641	1.706	1.748	1.692	1.687	1.709	1.731
3.A.2 - Manure Management	640	661	678	661	661	665	678
3.C - Aggregate sources and non-CO2 emissions sources on land	1.045	1.042	985	1.028	1.013	1.032	1.021
3.C.1 - Burning	8	9	10	10	9	9	8
3.C.2 - Liming	0	0	0	0	0	0	0
3.C.3 - Urea application	27	13	21	8	1	4	0
3.C.4 - Direct N2O Emissions from managed soils	650	672	613	664	657	665	655
3.C.5 - Indirect N2O Emissions from managed soils	260	240	231	240	239	242	246
3.C.6 - Indirect N2O Emissions from manure management	99	108	110	108	108	113	111
4 - Waste	1.046	1.090	1.101	1.142	1.182	1.221	1.258
4.A - Solid Waste Disposal	549	587	624	660	695	0	762
4.D - Wastewater Treatment and Discharge	496	503	477	482	487	492	496